

2014 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–4

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 5–15

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

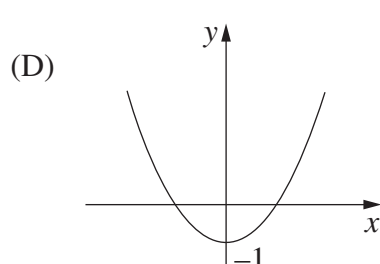
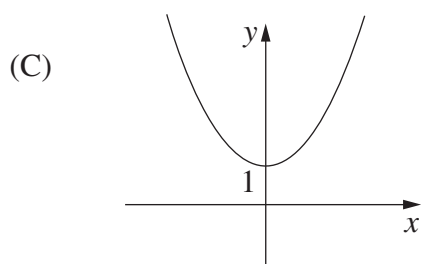
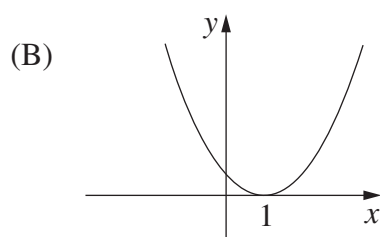
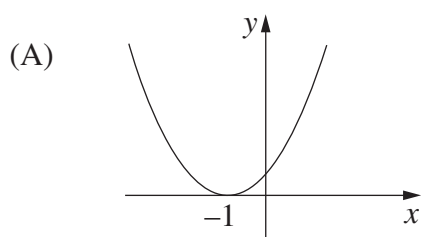
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What is the value of $\frac{\pi^2}{6}$, correct to 3 significant figures?

- (A) 1.64
- (B) 1.65
- (C) 1.644
- (D) 1.645

2 Which graph best represents $y = (x - 1)^2$?



3 What is the solution to the equation $\log_2(x - 1) = 8$?

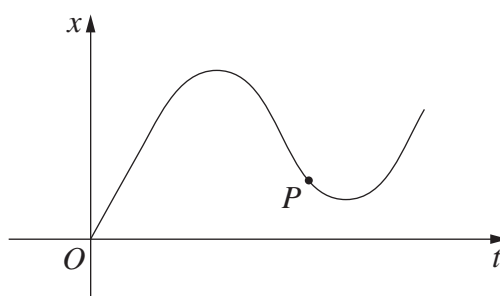
- (A) 4
- (B) 17
- (C) 65
- (D) 257

- 4 Which expression is equal to $\int e^{2x} dx$?
- (A) $e^{2x} + c$
- (B) $2e^{2x} + c$
- (C) $\frac{e^{2x}}{2} + c$
- (D) $\frac{e^{2x+1}}{2x+1} + c$
- 5 Which equation represents the line perpendicular to $2x - 3y = 8$, passing through the point $(2, 0)$?
- (A) $3x + 2y = 4$
- (B) $3x + 2y = 6$
- (C) $3x - 2y = -4$
- (D) $3x - 2y = 6$
- 6 Which expression is a factorisation of $8x^3 + 27$?
- (A) $(2x - 3)(4x^2 + 12x - 9)$
- (B) $(2x + 3)(4x^2 - 12x + 9)$
- (C) $(2x - 3)(4x^2 + 6x - 9)$
- (D) $(2x + 3)(4x^2 - 6x + 9)$
- 7 How many solutions of the equation $(\sin x - 1)(\tan x + 2) = 0$ lie between 0 and 2π ?
- (A) 1
- (B) 2
- (C) 3
- (D) 4

8 Which expression is a term of the geometric series $3x - 6x^2 + 12x^3 - \dots$?

- (A) $3072x^{10}$
- (B) $-3072x^{10}$
- (C) $3072x^{11}$
- (D) $-3072x^{11}$

9 The graph shows the displacement x of a particle moving along a straight line as a function of time t .



Which statement describes the motion of the particle at the point P ?

- (A) The velocity is negative and the acceleration is positive.
 - (B) The velocity is negative and the acceleration is negative.
 - (C) The velocity is positive and the acceleration is positive.
 - (D) The velocity is positive and the acceleration is negative.
- 10 Three runners compete in a race. The probabilities that the three runners finish the race in under 10 seconds are $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{2}{5}$ respectively.

What is the probability that at least one of the three runners will finish the race in under 10 seconds?

- (A) $\frac{1}{60}$
- (B) $\frac{37}{60}$
- (C) $\frac{3}{8}$
- (D) $\frac{5}{8}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

(a) Rationalise the denominator of $\frac{1}{\sqrt{5}-2}$. **2**

(b) Factorise $3x^2 + x - 2$. **2**

(c) Differentiate $\frac{x^3}{x+1}$. **2**

(d) Find $\int \frac{1}{(x+3)^2} dx$. **2**

(e) Evaluate $\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx$. **3**

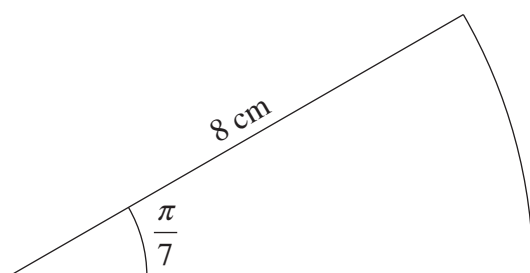
Question 11 continues on page 6

Question 11 (continued)

- (f) The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x - 5$. **2**
The curve passes through the point $(2, 3)$.

Find the equation of the curve.

- (g) The angle of a sector in a circle of radius 8 cm is $\frac{\pi}{7}$ radians, as shown in the diagram. **2**
The diagram shows a sector of a circle with a radius of 8 cm and a central angle of $\frac{\pi}{7}$ radians.



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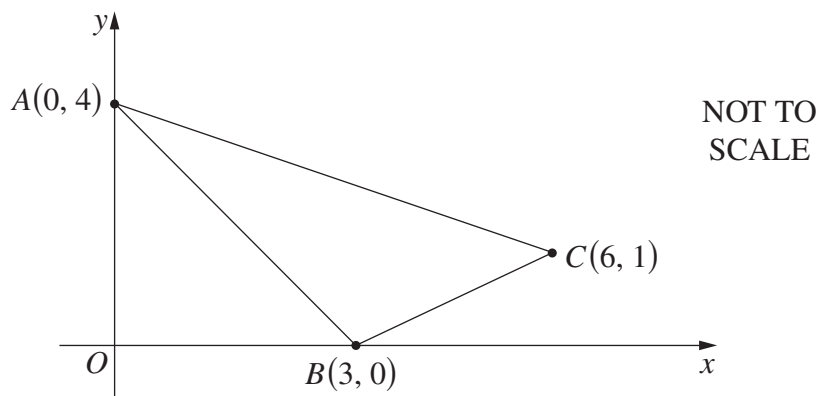
Find the exact value of the perimeter of the sector.

End of Question 11

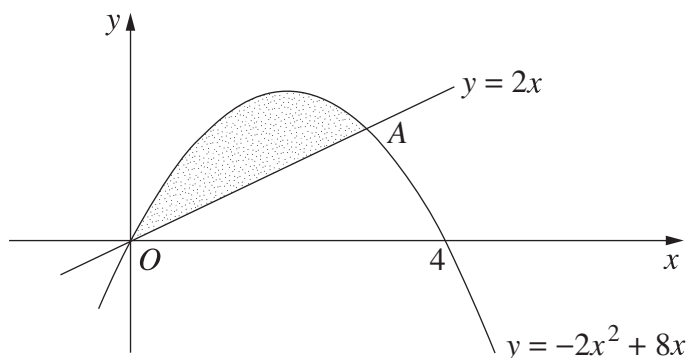
Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Evaluate the arithmetic series $2 + 5 + 8 + 11 + \dots + 1094$. **2**

- (b) The points $A(0, 4)$, $B(3, 0)$ and $C(6, 1)$ form a triangle, as shown in the diagram.



- (i) Show that the equation of AC is $x + 2y - 8 = 0$. **2**
- (ii) Find the perpendicular distance from B to AC . **2**
- (iii) Hence, or otherwise, find the area of $\triangle ABC$. **2**
- (c) A packet of lollies contains 5 red lollies and 14 green lollies. Two lollies are selected at random without replacement.
- (i) Draw a tree diagram to show the possible outcomes. Include the probability on each branch. **2**
- (ii) What is the probability that the two lollies are of different colours? **1**
- (d) The parabola $y = -2x^2 + 8x$ and the line $y = 2x$ intersect at the origin and at the point A .



- (i) Find the x -coordinate of the point A . **1**
- (ii) Calculate the area enclosed by the parabola and the line. **3**

Question 13 (15 marks) Use the Question 13 Writing Booklet.

(a) (i) Differentiate $3 + \sin 2x$. **1**

(ii) Hence, or otherwise, find $\int \frac{\cos 2x}{3 + \sin 2x} dx$. **2**

(b) A quantity of radioactive material decays according to the equation

$$\frac{dM}{dt} = -kM,$$

where M is the mass of the material in kg, t is the time in years and k is a constant.

(i) Show that $M = Ae^{-kt}$ is a solution to the equation, where A is a constant. **1**

(ii) The time for half of the material to decay is 300 years. If the initial amount of material is 20 kg, find the amount remaining after 1000 years. **3**

(c) The displacement of a particle moving along the x -axis is given by

$$x = t - \frac{1}{1+t},$$

where x is the displacement from the origin in metres, t is the time in seconds, and $t \geq 0$.

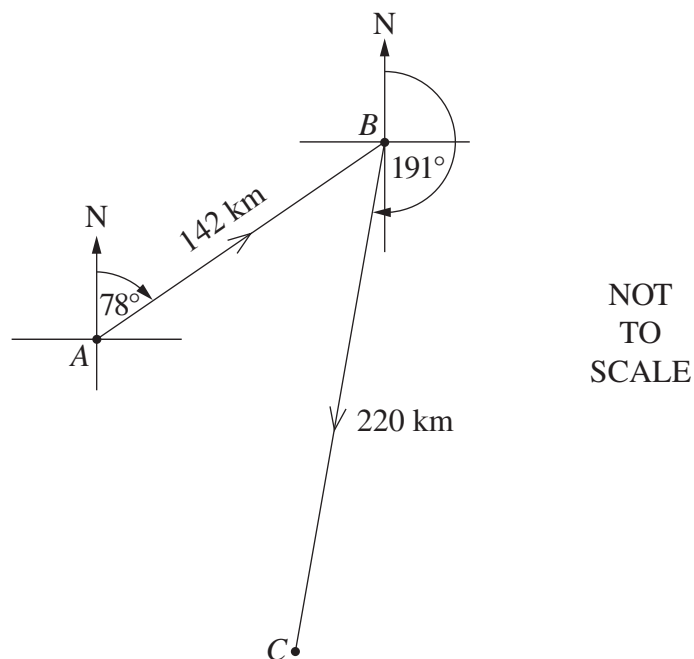
(i) Show that the acceleration of the particle is always negative. **2**

(ii) What value does the velocity approach as t increases indefinitely? **1**

Question 13 continues on page 9

Question 13 (continued)

- (d) Chris leaves island A in a boat and sails 142 km on a bearing of 078° to island B . Chris then sails on a bearing of 191° for 220 km to island C , as shown in the diagram.

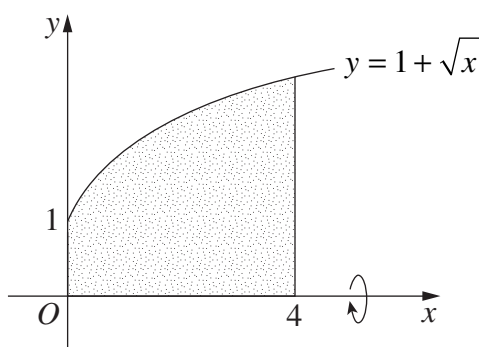


- (i) Show that the distance from island C to island A is approximately 210 km. **2**
- (ii) Chris wants to sail from island C directly to island A . On what bearing should Chris sail? Give your answer correct to the nearest degree. **3**

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) Find the coordinates of the stationary point on the graph $y = e^x - ex$, and determine its nature. **3**
- (b) The roots of the quadratic equation $2x^2 + 8x + k = 0$ are α and β .
- (i) Find the value of $\alpha + \beta$. **1**
- (ii) Given that $\alpha^2\beta + \alpha\beta^2 = 6$, find the value of k . **2**
- (c) The region bounded by the curve $y = 1 + \sqrt{x}$ and the x -axis between $x = 0$ and $x = 4$ is rotated about the x -axis to form a solid. **3**



Find the volume of the solid.

- (d) At the beginning of every 8-hour period, a patient is given 10 mL of a particular drug.
- During each of these 8-hour periods, the patient's body partially breaks down the drug. Only $\frac{1}{3}$ of the total amount of the drug present in the patient's body at the beginning of each 8-hour period remains at the end of that period.
- (i) How much of the drug is in the patient's body immediately after the second dose is given? **1**
- (ii) Show that the total amount of the drug in the patient's body never exceeds 15 mL. **2**

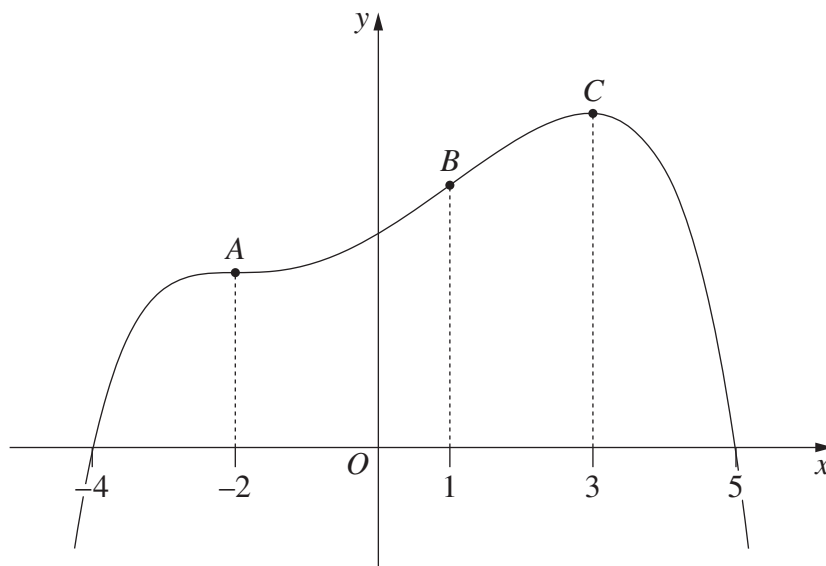
Question 14 continues on page 11

Question 14 (continued)

- (e) The diagram shows the graph of a function $f(x)$.

3

The graph has a horizontal point of inflexion at A , a point of inflexion at B and a maximum turning point at C .



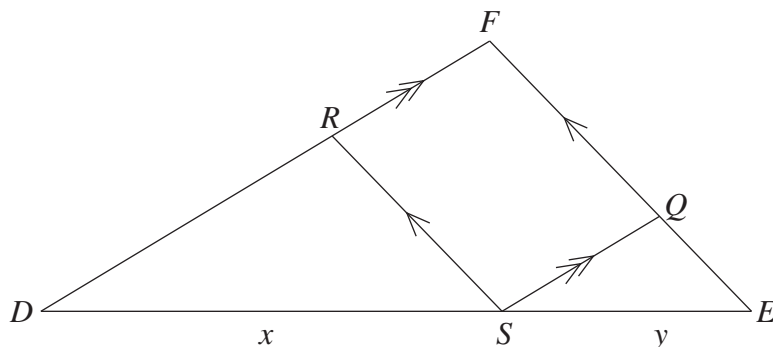
Sketch the graph of the derivative $f'(x)$.

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) Find all solutions of $2 \sin^2 x + \cos x - 2 = 0$, where $0 \leq x \leq 2\pi$. **3**

- (b) In $\triangle DEF$, a point S is chosen on the side DE . The length of DS is x , and the length of ES is y . The line through S parallel to DF meets EF at Q . The line through S parallel to EF meets DF at R .



The area of $\triangle DEF$ is A . The areas of $\triangle DSR$ and $\triangle SEQ$ are A_1 and A_2 respectively.

- (i) Show that $\triangle DEF$ is similar to $\triangle DSR$. **2**
- (ii) Explain why $\frac{DR}{DF} = \frac{x}{x+y}$. **1**
- (iii) Show that $\sqrt{\frac{A_1}{A}} = \frac{x}{x+y}$. **2**
- (iv) Using the result from part (iii) and a similar expression for $\sqrt{\frac{A_2}{A}}$, **2**
deduce that $\sqrt{A} = \sqrt{A_1} + \sqrt{A_2}$.

Question 15 continues on page 13

Question 15 (continued)

(c) The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P .

- | | |
|---|----------|
| (i) Sketch the line and the curve on one diagram. | 1 |
| (ii) Find the coordinates of P . | 3 |
| (iii) Find the value of m . | 1 |

End of Question 15

Please turn over

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) Use Simpson's Rule with five function values to show that

3

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sec x \, dx \approx \frac{\pi}{9} \left(3 + \frac{8}{\sqrt{3}} \right).$$

- (b) At the start of a month, Jo opens a bank account and makes a deposit of \$500. At the start of each subsequent month, Jo makes a deposit which is 1% more than the previous deposit.

At the end of every month, the bank pays interest of 0.3% (per month) on the balance of the account.

- (i) Explain why the balance of the account at the end of the second month is

2

$$\$500(1.003)^2 + \$500(1.01)(1.003).$$

- (ii) Find the balance of the account at the end of the 60th month, correct to the nearest dollar.

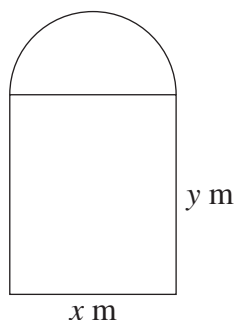
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Question 16 continues on page 15

Question 16 (continued)

- (c) The diagram shows a window consisting of two sections. The top section is a semicircle of diameter x m. The bottom section is a rectangle of width x m and height y m.

The entire frame of the window, including the piece that separates the two sections, is made using 10 m of thin metal.



The semicircular section is made of coloured glass and the rectangular section is made of clear glass.

Under test conditions the amount of light coming through one square metre of the coloured glass is 1 unit and the amount of light coming through one square metre of the clear glass is 3 units.

The total amount of light coming through the window under test conditions is L units.

- (i) Show that $y = 5 - x \left(1 + \frac{\pi}{4} \right)$. **2**
- (ii) Show that $L = 15x - x^2 \left(3 + \frac{5\pi}{8} \right)$. **2**
- (iii) Find the values of x and y that maximise the amount of light coming through the window under test conditions. **3**

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

2014 HSC Mathematics Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	A
2	B
3	D
4	C
5	B
6	D
7	B
8	C
9	A
10	D

Section II

Question 11 (a)

Criteria	Marks
• Provides a correct answer	2
• Attempts to use $\sqrt{5} + 2$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\frac{1}{\sqrt{5}-2} &= \frac{\sqrt{5}+2}{(\sqrt{5}-2)(\sqrt{5}+2)} \\ &= \frac{\sqrt{5}+2}{5-4} \\ &= \sqrt{5}+2\end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides a correct answer	2
• Finds product of the form $(3x+a)(x+b)$, or equivalent merit	1

Sample answer:

$$3x^2 + x - 2 = (3x - 2)(x + 1)$$

Question 11 (c)

Criteria	Marks
• Finds correct derivative	2
• Attempts to use quotient rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}y &= \frac{x^3}{x+1} \\ \frac{dy}{dx} &= \frac{3x^2(x+1) - x^3}{(x+1)^2} = \frac{(2x+3)x^2}{(x+1)^2}\end{aligned}$$

Question 11 (d)

Criteria	Marks
• Finds correct primitive	2
• Writes integral in the form $\int (x+3)^{-2} dx$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\int \frac{1}{(x+3)^2} dx &= \int (x+3)^{-2} dx \\ &= -(x+3)^{-1} + C\end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides a correct solution	3
• Finds a correct primitive	2
• Finds a primitive of the form $a \cos \frac{x}{2}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx &= \left[-2 \cos \frac{x}{2} \right]_0^{\frac{\pi}{2}} \\ &= -2 \cos \frac{\pi}{4} + 2 \cos 0 \\ &= -\sqrt{2} + 2 \\ &= 2 - \sqrt{2}\end{aligned}$$

Question 11 (f)

Criteria	Marks
• Finds the correct equation for the curve	2
• Finds a correct primitive for $4x - 5$, or equivalent merit	1

Sample answer:

$$f'(x) = 4x - 5$$

$$f(x) = 2x^2 - 5x + C$$

$$3 = f(2) = 8 - 10 + C \text{ (using that the curve passes through (2,3))}$$

$$\therefore C = 5$$

$$f(x) = 2x^2 - 5x + 5$$

Question 11 (g)

Criteria	Marks
• Finds the correct perimeter	2
• Finds the length of the circular arc, or equivalent merit	1

Sample answer:

Arc length:

$$l = r\theta$$

$$= 8 \times \frac{\pi}{7} = \frac{8\pi}{7}$$

$$\text{Perimeter} = \left(8 + 8 + \frac{8\pi}{7} \right) \text{ cm}$$

$$= \left(16 + \frac{8\pi}{7} \right) \text{ cm}$$

Question 12 (a)

Criteria	Marks
• Finds the correct value	2
• Finds the number of terms in the series, or equivalent merit	1

Sample answer:

$$2 + 5 + 8 + \cdots + 1094$$

$$T_n = 2 + 3(n - 1)$$

$$1094 = 2 + 3(n - 1)$$

$$1092 = 3(n - 1)$$

$$n = 364 + 1 = 365$$

$$S_n = \frac{n}{2}(a + l)$$

$$S_{365} = \frac{365}{2}(2 + 1094)$$

$$= 200020$$

Question 12 (b) (i)

Criteria	Marks
• Provides a correct solution	2
• Finds the slope of AC, or equivalent merit	1

Sample answer:

$$\frac{y - 4}{x - 0} = \frac{1 - 4}{6 - 0}$$

$$6y - 24 = -3x$$

$$3x + 6y - 24 = 0$$

$$x + 2y - 8 = 0$$

Question 12 (b) (ii)

Criteria	Marks
• Provides a correct solution	2
• Attempts to use an appropriate formula, or equivalent merit	1

Sample answer:

$$\text{Perpendicular distance} = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$\begin{aligned} \text{perpendicular distance} &= \frac{|3 + 2 \times 0 - 8|}{\sqrt{1^2 + 2^2}} \\ &= \frac{5}{\sqrt{5}} \\ &= \sqrt{5} \end{aligned}$$

Question 12 (b) (iii)

Criteria	Marks
• Finds the correct area	2
• Finds the length of AC, or equivalent merit	1

Sample answer:

$$\text{Area} = \frac{1}{2} \text{base} \times \text{height}$$

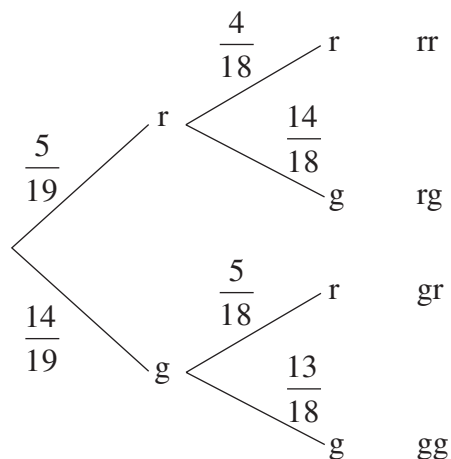
$$\begin{aligned} \text{base} = AC &= \sqrt{(6-0)^2 + (1-4)^2} \\ &= \sqrt{36+9} \\ &= \sqrt{45} \\ &= 3\sqrt{5} \end{aligned}$$

Height is $\sqrt{5}$ (from part (ii))

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 3\sqrt{5} \times \sqrt{5} \\ &= \frac{15}{2} \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Draws a correct diagram, including probability on each branch	2
• Draws a correct tree diagram	1

Sample answer:**Question 12 (c) (ii)**

Criteria	Marks
• Finds the correct probability	1

Sample answer:

$$\begin{aligned}
 P(rg) + P(gr) &= \frac{5}{19} \cdot \frac{14}{18} + \frac{14}{19} \cdot \frac{5}{18} \\
 &= \frac{70}{171}
 \end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Finds the correct x -coordinate	1

Sample answer:

y values are equal at A

$$2x = -2x^2 + 8x$$

$$2x^2 - 6x = 0$$

$$2x(x - 3) = 0$$

$$x = 0 \text{ or } x = 3$$

The x coordinate of point A is 3 (since $x = 0$ corresponds to point O).

Question 12 (d) (ii)

Criteria	Marks
• Finds the correct area	3
• Finds a correct expression for the area, and attempts to evaluate it	2
• Attempts to express the area as a difference, or equivalent merit	1

Sample answer:

Area is below the parabola but above line, so

$$\text{Area} = \int_0^3 (-2x^2 + 8x) - 2x \, dx$$

$$= \int_0^3 -2x^2 + 6x \, dx$$

$$= \left[-\frac{2x^3}{3} + 3x^2 \right]_0^3$$

$$= -18 + 27$$

$$= 9 \text{ units}^2$$

Question 13 (a) (i)

Criteria	Marks
• Finds the correct derivative	1

Sample answer:

$$y = 3 + \sin 2x$$

$$\frac{dy}{dx} = 2 \cos 2x$$

Question 13 (a) (ii)

Criteria	Marks
• Finds a correct primitive	2
• Finds a primitive of the form $a \log f(x)$, or equivalent merit	1

Sample answer:

$$\int \frac{\cos 2x}{3 + \sin 2x} dx = \frac{1}{2} \int \frac{2 \cos 2x}{3 + \sin 2x} dx$$

This is of the form $\frac{f'(x)}{f(x)}$ with $f(x) = 3 + \sin 2x$.

$$\text{Hence, } \int \frac{\cos 2x}{3 + \sin 2x} dx = \frac{1}{2} \ln(3 + \sin 2x) + c.$$

Question 13 (b) (i)

Criteria	Marks
• Provides a correct solution	1

Sample answer:

$$M = Ae^{-kt}$$

$$\begin{aligned}\frac{dM}{dt} &= -kAe^{-kt} \\ &= -kM\end{aligned}$$

Question 13 (b) (ii)

Criteria	Marks
• Provides a correct solution	3
• Finds the value of k , or equivalent merit	2
• Finds the value of A , or equivalent merit	1

Sample answer:When $t = 0$, $M = 20$

$$20 = Ae^{-k \cdot 0} = A$$

$$\therefore M = 20e^{-kt}$$

When $t = 300$, $M = 10$

$$10 = 20e^{-300k}$$

$$\frac{1}{2} = e^{-300k}$$

$$-300k = \log \frac{1}{2} = -\log 2$$

$$k = \frac{\log 2}{300}$$

When $t = 1000$

$$M = 20e^{-1000k}$$

$$= 20e^{-\frac{10}{3}\log 2}$$

$$= 1.984\dots$$

 $\therefore$ The amount remaining after 1000 years is approximately 1.98 kg

Question 13 (c) (i)

Criteria	Marks
• Provides a correct solution	2
• Finds $\frac{dx}{dt}$, or equivalent merit	1

Sample answer:

$$v = \frac{dx}{dt} = 1 + \frac{1}{(1+t)^2}$$

$$a = \frac{d^2x}{dt^2} = -\frac{2}{(1+t)^3}$$

$$< 0 \text{ as } (1+t)^3 > 0 \text{ for } t \geq 0.$$

Question 13 (c) (ii)

Criteria	Marks
• Finds the correct value	1

Sample answer:

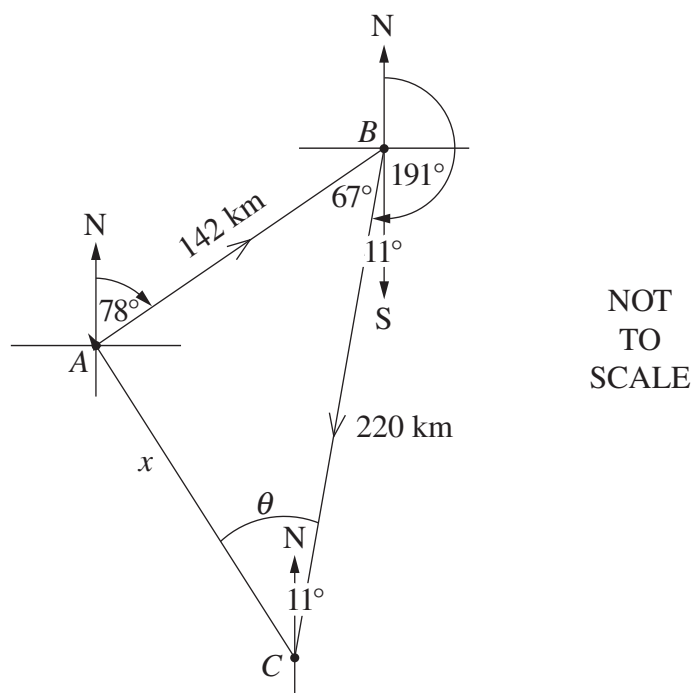
$$\lim_{t \rightarrow \infty} \left(1 + \frac{1}{(1+t)^2} \right)$$

$$= 1 \text{ since } \frac{1}{(1+t)^2} \rightarrow 0 \text{ as } t \rightarrow \infty$$

velocity $\rightarrow 1$ as t becomes large.

Question 13 (d) (i)

Criteria	Marks
• Provides a correct solution	2
• Attempts to apply the cosine rule, or equivalent merit	1

Sample answer:

$\angle ABS = 78^\circ$ (parallel lines, alternate angles are equal)

$\angle SBC = 191^\circ - 180^\circ = 11^\circ$

$\therefore \angle ABC = 78^\circ - 11^\circ = 67^\circ$

Using Cosine rule,

$$\begin{aligned} AC^2 &= 142^2 + 220^2 - 2 \times 142 \times 220 \times \cos 67^\circ \\ &= 44151.119 \dots \end{aligned}$$

$$\therefore AC = \sqrt{44151.19 \dots}$$

$$= 210.121 \dots$$

$$= 210 \text{ km (approximately)}$$

Question 13 (d) (ii)

Criteria	Marks
• Finds the correct bearing	3
• Finds angle $\angle ACB$ or $\angle CAB$, or equivalent merit	2
• Attempts to use the sine or cosine rule to find $\angle ACB$ or $\angle CAB$, or equivalent merit	1

Sample answer:

Using Sine Rule,

$$\frac{\sin 67^\circ}{AC} = \frac{\sin \theta}{142}$$

$$\begin{aligned}\sin \theta &= \frac{\sin 67^\circ \times 142}{210.12 \dots} \\ &= 0.6220 \dots\end{aligned}$$

$$\begin{aligned}\therefore \theta &= \sin^{-1}(0.622 \dots) \\ &= 38^\circ 28'\end{aligned}$$

$$\angle NCB = 11^\circ \text{ (parallel lines, alternate to } \angle SBC \text{)}$$

$$\begin{aligned}\therefore \angle NCA &= 38^\circ 28' - 11^\circ \\ &= 27^\circ 28'\end{aligned}$$

$$\begin{aligned}\therefore \text{Bearing} &= 360^\circ - 27^\circ 28' \\ &= 332^\circ 32' \\ &\approx 333^\circ\end{aligned}$$

Question 14 (a)

Criteria	Marks
• Provides a correct solution	3
• Finds the coordinates of the stationary point, or equivalent merit	2
• Finds the correct derivative, or equivalent merit	1

Sample answer:

$$y = e^x - ex$$

$$\frac{dy}{dx} = e^x - e$$

At stationary point $\frac{dy}{dx} = 0$

$$0 = e^x - e$$

$$e^x = e$$

$$x = 1$$

$$y = e - e \times 1 = 0$$

$(1,0)$ is stationary point.

Determine nature:

$$\frac{d^2y}{dx^2} = e^x$$

at $x = 1$, $\frac{d^2y}{dx^2} = e > 0$ (cu)

So $(1,0)$ is a minimum stationary point.

Question 14 (b) (i)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$\alpha + \beta = \frac{-8}{2} = -4$$

Question 14 (b) (ii)

Criteria	Marks
• Provides a correct solution	2
• Finds the value of $\alpha\beta$, or equivalent merit	1

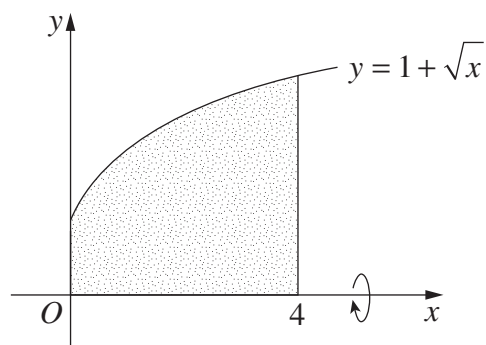
Sample answer:

$$\begin{aligned} 6 &= \alpha^2\beta + \alpha\beta^2 \\ &= \alpha\beta(\alpha + \beta) \\ &= \alpha\beta(-4) \\ \alpha\beta &= \frac{-3}{2} \end{aligned}$$

$$\begin{aligned} \text{But, } \alpha\beta &= \frac{k}{2} \\ \therefore \frac{k}{2} &= -\frac{3}{2} \\ k &= -3 \end{aligned}$$

Question 14 (c)

Criteria	Marks
• Provides a correct solution	3
• Writes the volume as an integral, and attempts to find a primitive, or equivalent merit	2
• Attempts to find an integral for the volume, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 V &= \pi \int_0^4 y^2 dx \\
 &= \pi \int_0^4 (1 + \sqrt{x})^2 dx \\
 &= \pi \int_0^4 (1 + 2\sqrt{x} + x) dx \\
 &= \pi \left[x + 2 \times \frac{2}{3} x^{\frac{3}{2}} + \frac{x^2}{2} \right]_0^4 \\
 &= \pi \left[\left(4 + \frac{4}{3} \times 4^{\frac{3}{2}} + \frac{4^2}{2} \right) - 0 \right] \\
 &= \pi \times \frac{68}{3}
 \end{aligned}$$

$$\text{Volume} = \frac{68\pi}{3} \text{ unit}^3$$

Question 14 (d) (i)

Criteria	Marks
• Provides the correct answer	1

Sample answer:

$$10 + \frac{1}{3}(10)$$

$$= 10 + \frac{10}{3}$$

Question 14 (d) (ii)

Criteria	Marks
• Provides a correct solution	2
• Recognises a geometric series, or equivalent merit	1

Sample answer:

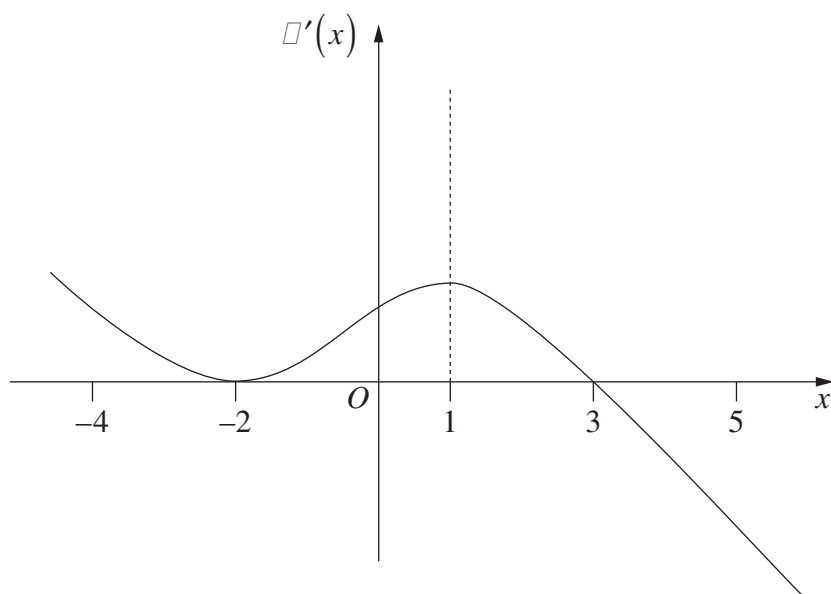
Amount remaining after many doses is $10 + \frac{10}{3} + 10\left(\frac{1}{3}\right)^2 + \dots$

Limiting sum is $\frac{10}{1 - \frac{1}{3}} = 15$. Amount in body is no more than 15 mL.

Question 14 (e)

Criteria	Marks
• Correctly sketches the graph	3
• Sketches graph of the derivative correct at $x = 3$ and $x = -2$, or equivalent merit	2
• Sketches a graph of the derivative correct at $x = 3$, or equivalent merit	1

Sample answer:



Question 15 (a)

Criteria	Marks
• Provides a correct solution	3
• Finds two correct solutions of the equation	2
• Obtains a quadratic in $\cos x$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 2\sin^2 x + \cos x - 2 &= 0 \\
 2(1 - \cos^2 x) + \cos x - 2 &= 0 \\
 -2\cos^2 x + \cos x &= 0 \\
 \cos x(1 - 2\cos x) &= 0 \\
 \cos x = 0 &\quad \text{or} \quad 1 - 2\cos x = 0 \\
 x = \frac{\pi}{2} \text{ or } \frac{3\pi}{2} &\quad \cos x = \frac{1}{2} \\
 &\quad x = \frac{\pi}{3} \text{ or } \frac{5\pi}{3} \\
 \text{Solutions are } \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{3}, \frac{5\pi}{3}.
 \end{aligned}$$

Question 15 (b) (i)

Criteria	Marks
• Provides a correct proof	2
• Establishes one pair of equal angles, or equivalent merit	1

Sample answer:

$RS \parallel FE$ so $\angle DSR = \angle DEF$ (corresponding angles)

Also $\angle RDS = \angle FDE$ (common angle)

so $\triangle DEF$ and $\triangle DSR$ are similar (equiangular)

Question 15 (b) (ii)

Criteria	Marks
• Provides a correct explanation	1

Sample answer:

$$\begin{aligned}\frac{DR}{DF} &= \frac{DS}{DE} \\ &= \frac{DS}{DS + SE} = \frac{x}{x + y}\end{aligned}$$

Question 15 (b) (iii)

Criteria	Marks
• Provides a correct solution	2
• Uses expressions for the area of triangles, or equivalent merit	1

Sample answer:

$$A_1 = \frac{1}{2} DR \times DS \sin D \quad (\text{using area formula: } A = \frac{1}{2} ab \sin C)$$

$$A = \frac{1}{2} DF \times DE \sin D$$

$$\frac{A_1}{A} = \frac{DR}{DF} \times \frac{DS}{DE}$$

$$= \frac{x}{x + y} \times \frac{x}{x + y} = \left(\frac{x}{x + y} \right)^2$$

$$\sqrt{\frac{A_1}{A}} = \frac{x}{x + y}$$

Question 15 (b) (iv)

Criteria	Marks
• Provides a correct solution	2
• Observes that $\triangle DEF$ and $\triangle SEQ$ are similar, or equivalent merit	1

Sample answer:

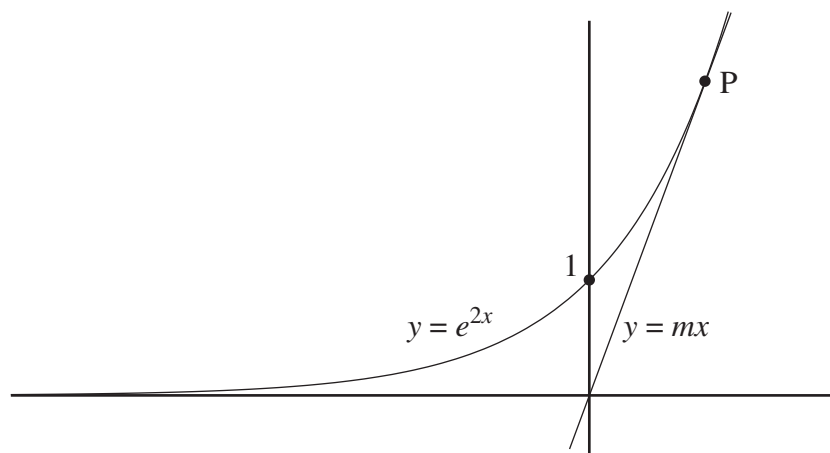
$\triangle SEQ$ is similar to $\triangle DEF$ and as in (iii) $\sqrt{\frac{A_2}{A}} = \frac{y}{x+y}$.

$$\text{Hence } \sqrt{\frac{A_1}{A}} + \sqrt{\frac{A_2}{A}} = \frac{x}{x+y} + \frac{y}{x+y}$$

$$\text{Thus } \sqrt{A_1} + \sqrt{A_2} = \sqrt{A} \quad = 1.$$

Question 15 (c) (i)

Criteria	Marks
• Draws a correct sketch	1

Sample answer:

Question 15 (c) (ii)

Criteria	Marks
• Finds the coordinates of P	3
• Attempts to solve relevant simultaneous equations, or equivalent merit	2
• Attempts to use the fact that P lies on both curves, or equivalent merit	1

Sample answer:

At P the y-values of the line and the curve coincide:

$$y = mx = e^{2x} \quad (1)$$

At P the slopes of the line and the tangent to the curve coincide:

$$m = \frac{d}{dx}e^{2x} = 2e^{2x} \quad (2)$$

Substitute (1) into (2):

$$m = 2mx$$

$$1 = 2x$$

$$x = \frac{1}{2}$$

P lies on the curve, so $y = e^{2 \times \frac{1}{2}} = e$.

The coordinates of P are $\left(\frac{1}{2}, e\right)$.

Question 15 (c) (iii)

Criteria	Marks
• Finds the value of m	1

Sample answer:

At P , $mx = e^{2x}$

Using $x = \frac{1}{2}$ from (ii)

$$\begin{aligned} m &= 2e^{2 \times \frac{1}{2}} \\ &= 2e \end{aligned}$$

Question 16 (a)

Criteria	Marks
• Provides a correct solution	3
• Correctly evaluates function values and uses them in Simpson's rule, or equivalent merit	2
• Identifies suitable function values, or equivalent merit	1

Sample answer:

Consider the following table of values of $\sec x$.

x	$-\frac{\pi}{3}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$
$\cos x$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$
$\sec x$	2	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	2

Using Simpson's rule

$$\begin{aligned}
 \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sec x \, dx &\approx \frac{1}{3} \frac{\pi}{6} \left(\sec\left(-\frac{\pi}{3}\right) + 4\sec\left(-\frac{\pi}{6}\right) + 2\sec 0 + 4\sec\left(\frac{\pi}{6}\right) + \sec\left(\frac{\pi}{3}\right) \right) \\
 &= \frac{\pi}{18} \left(2 + \frac{8}{\sqrt{3}} + 2 + \frac{8}{\sqrt{3}} + 2 \right) \\
 &= \frac{\pi}{18} \left(6 + \frac{16}{\sqrt{3}} \right) \\
 &= \frac{\pi}{9} \left(3 + \frac{8}{\sqrt{3}} \right)
 \end{aligned}$$

Question 16 (b) (i)

Criteria	Marks
• Provides a complete explanation	2
• Explains the meaning of one term, or equivalent merit	1

Sample answer:

Start	1st month	500
End	1st month (add interest)	$500 (1.003)$
Start	2nd month (add principal)	$500 (1.003) + 500 (1.01)$
End	2nd month (add interest)	$[500 (1.003) + 500 (1.01)] 1.003$ $= 500 (1.003)^2 + 500 (1.01) (1.003)$

Question 16 (b) (ii)

Criteria	Marks
• Provides a correct solution	3
• Finds a relevant geometric series, or equivalent merit	2
• Attempts to find an expression for the balance, or equivalent merit	1

Sample answer:

At the end of the 3rd month we have

$$500 (1.003)^3 + 500 (1.01) (1.003)^2 + 500 (1.01)^2 (1.003)$$

Following the pattern, at end of 60th month

$$\begin{aligned}
 B &= 500 (1.003)^{60} + 500 (1.01) (1.003)^{59} + \dots + 500 (1.01)^{59} (1.003) \\
 &= 500 (1.003)^{60} + 500 (1.003)^{60} \frac{1.01}{1.003} + \dots + 500 (1.003)^{60} \left(\frac{1.01}{1.003} \right)^{59} \\
 &= 500 (1.003)^{60} \left(1 + \frac{1.01}{1.003} + \left(\frac{1.01}{1.003} \right)^2 + \dots + \left(\frac{1.01}{1.003} \right)^{59} \right) \\
 &= 500 (1.003)^{60} \frac{\left(\frac{1.01}{1.003} \right)^{60} - 1}{\frac{1.01}{1.003} - 1} \\
 &= 44404.378 \dots
 \end{aligned}$$

Question 16 (c) (i)

Criteria	Marks
• Provides a correct solution	2
• Finds an expression for the length of the frame, or equivalent merit	1

Sample answer:

$$\text{Length of semi-circle} = \pi \times \frac{x}{2} = \frac{\pi x}{2}$$

$$\text{Perimeter of rectangle} = 2x + 2y$$

Length of material used:

$$10 = 2x + 2y + \frac{\pi x}{2}$$

$$5 = x + y + \frac{\pi x}{4}$$

$$y = 5 - x \left(1 + \frac{\pi}{4} \right)$$

Question 16 (c) (ii)

Criteria	Marks
• Provides a correct solution	2
• Attempts to relate area of glass and the amount of light, or equivalent merit	1

Sample answer:

$$\text{Area of semicircle} = \frac{1}{2}\pi\left(\frac{x}{2}\right)^2 \text{ m}^2 = \frac{\pi x^2}{8} \text{ m}^2$$

$$\text{Area of rectangle} = xy \text{ m}^2$$

$$\text{Amount of light coming through} = \left[1\left(\frac{\pi x^2}{8}\right) + 3(xy)\right] \text{ units}$$

$$\text{ie } L = 3xy + \frac{\pi x^2}{8}$$

$$= 3x\left(5 - x - \frac{\pi x}{4}\right) + \frac{\pi x^2}{8}$$

$$= 15x - 3x^2 - \frac{3\pi x^2}{4} + \frac{\pi x^2}{8}$$

$$= 15x - x^2\left(3 + \frac{5\pi}{8}\right)$$

Question 16 (c) (iii)

Criteria	Marks
• Provides a correct solution	3
• Justifies the x value is a maximum, or finds y , or equivalent merit	2
• Finds the x value of the stationary point, or equivalent merit	1

Sample answer:

At maximum stationary point, $\frac{dL}{dx} = 0$

$$\frac{dL}{dx} = 15 - 2x\left(3 + \frac{5\pi}{8}\right) = 0$$

$$15 = 2x\left(3 + \frac{5\pi}{8}\right)$$

$$x = \frac{15}{2\left(3 + \frac{5\pi}{8}\right)}$$

$$\frac{d^2L}{dx^2} = -2\left(3 + \frac{5\pi}{8}\right) < 0$$

$$\text{so } x = \frac{15}{2\left(3 + \frac{5\pi}{8}\right)} \text{ gives maximum.}$$

Exact value $y = 5 - \left(\frac{15}{2\left(3 + \frac{5\pi}{8}\right)}\right)\left(1 + \frac{\pi}{4}\right)$

Decimal approximation for $y \doteq 2.30$
 $x \doteq 1.51$

Mathematics

2014 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.1	P3
2	1	4.2	P5
3	1	12.3	H3
4	1	12.5	H3, H5
5	1	6.2	P4
6	1	1.3	P3
7	1	5.2	H5
8	1	7.3	H5
9	1	10.1, 10.3, 14.3	H5, H7
10	1	3.3	H5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	1.1	P3
11 (b)	2	1.3	P3
11 (c)	2	8.8	P7, P8
11 (d)	2	11.2	H5, H8
11 (e)	3	13.7	H5, H8
11 (f)	2	10.8, 8.6	P6, H5, P8
11 (g)	2	13.1	H5
12 (a)	2	7.1	H5
12 (b) (i)	2	6.2	P4
12 (b) (ii)	2	6.5	P4
12 (b) (iii)	2	6.8	H2
12 (c) (i)	2	3.3	H5
12 (c) (ii)	1	3.3	H5
12 (d) (i)	1	1.4	P4
12 (d) (ii)	3	11.4	H8

Question	Marks	Content	Syllabus outcomes
13 (a) (i)	1	13.7	H5
13 (a) (ii)	2	11.2, 12.5, 13.7	H3, H5
13 (b) (i)	1	14.2	H3, H4
13 (b) (ii)	3	14.2	H3, H4
13 (c) (i)	2	14.3	H5
13 (c) (ii)	1	4.2, 14.3	H5, P4
13 (d) (i)	2	5.5	H2, H5, P4
13 (d) (ii)	3	5.4, 5.5	H2, H5, P4
14 (a)	3	10.2, 10.4, 12.4	H5
14 (b) (i)	1	9.2	P4
14 (b) (ii)	2	1.4, 9.2	P4
14 (c)	3	11.4	H8, H9
14 (d) (i)	1	7.2	H5
14 (d) (ii)	2	7.3	H5
14 (e)	3	10.2, 10.4	H7
15 (a)	3	5.2, 13.2	P4, H5
15 (b) (i)	2	2.3	P4, H9
15 (b) (ii)	1	2.5	H5, H9
15 (b) (iii)	2	2.5	H5, H9
15 (b) (iv)	2	2.5	H2, H5, H9
15 (c) (i)	1	4.2, 6.1, 12.3	P5, H3, H5
15 (c) (ii)	3	1.4, 10.1	H3, H5, H9
15 (c) (iii)	1	10.1	H3
16 (a)	3	11.3, 13.2	H5, H8
16 (b) (i)	2	7.5	H2, H5, H9
16 (b) (ii)	3	7.5	H5
16 (c) (i)	2	10.6	H4, H5, H9
16 (c) (ii)	2	10.6	H4, H5, H9
16 (c) (iii)	3	10.6	H5